Kernix 0.8.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

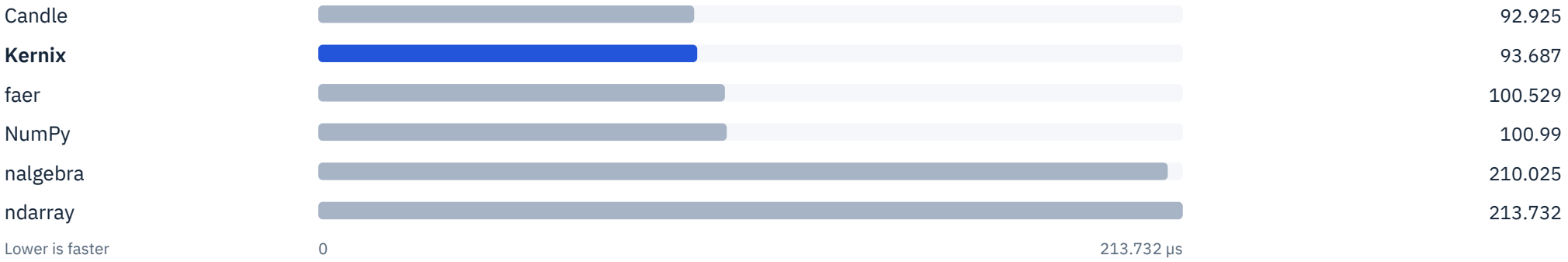
21 workloads

88 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

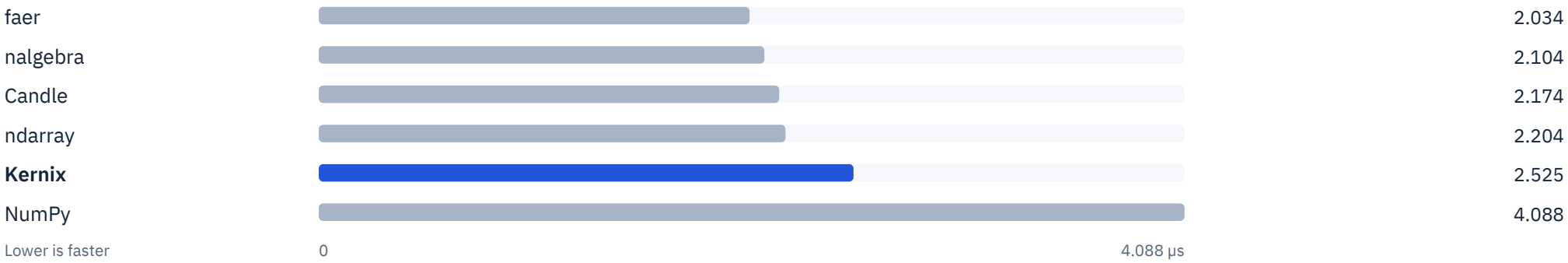
Kernix 0.8.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

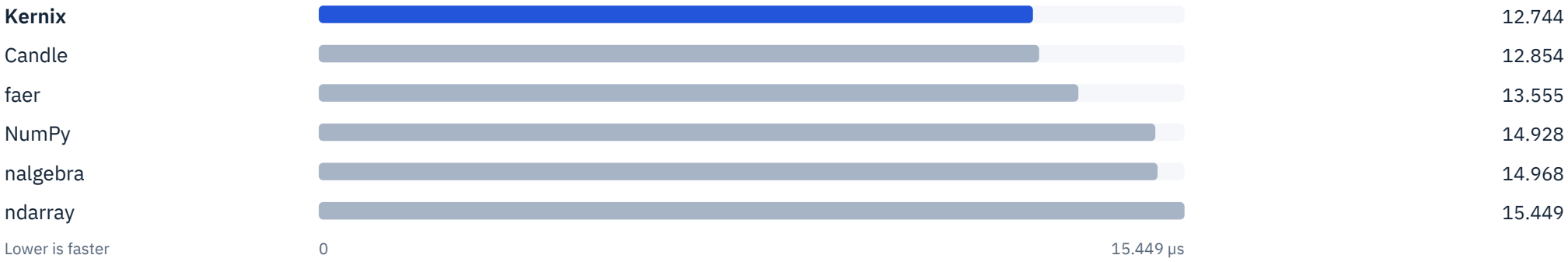
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



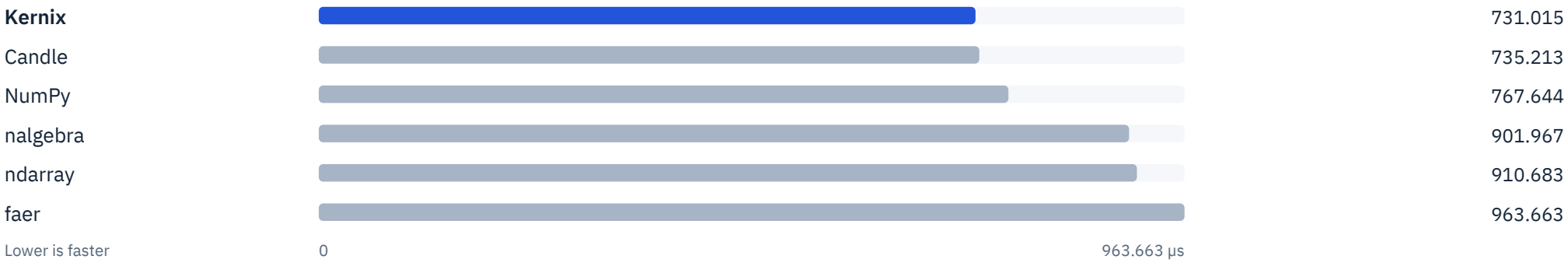
Kernix 0.8.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

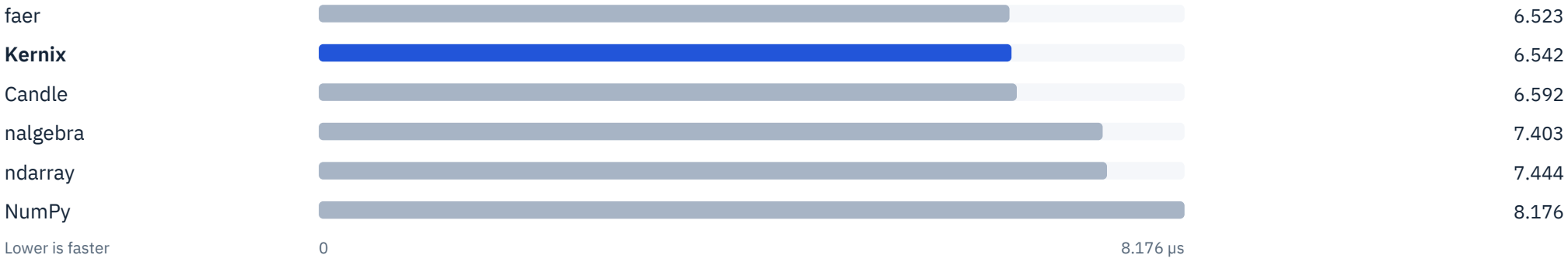
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



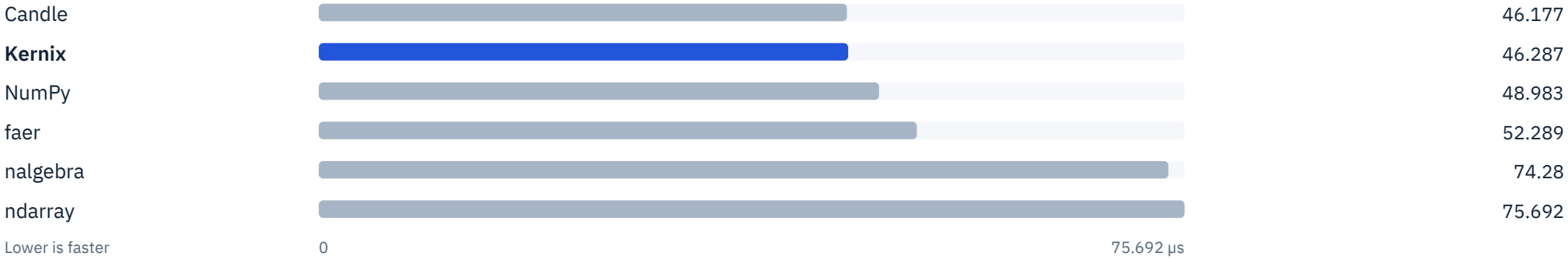
Kernix 0.8.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

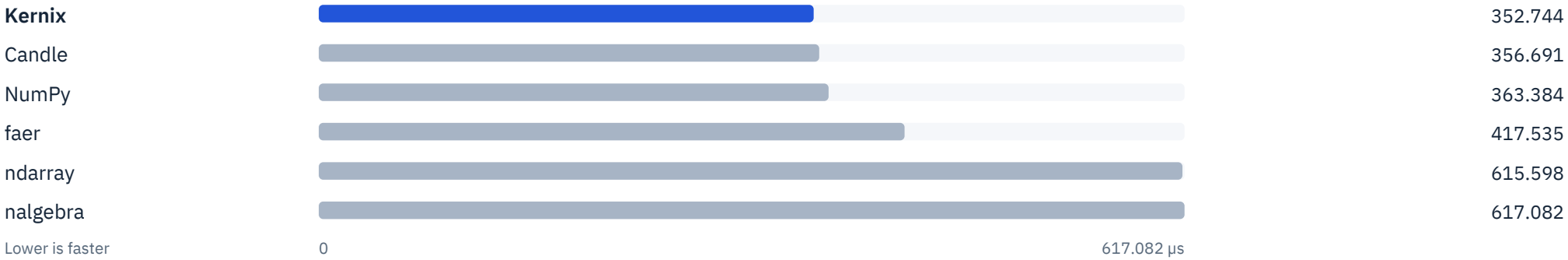
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.8.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.8.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

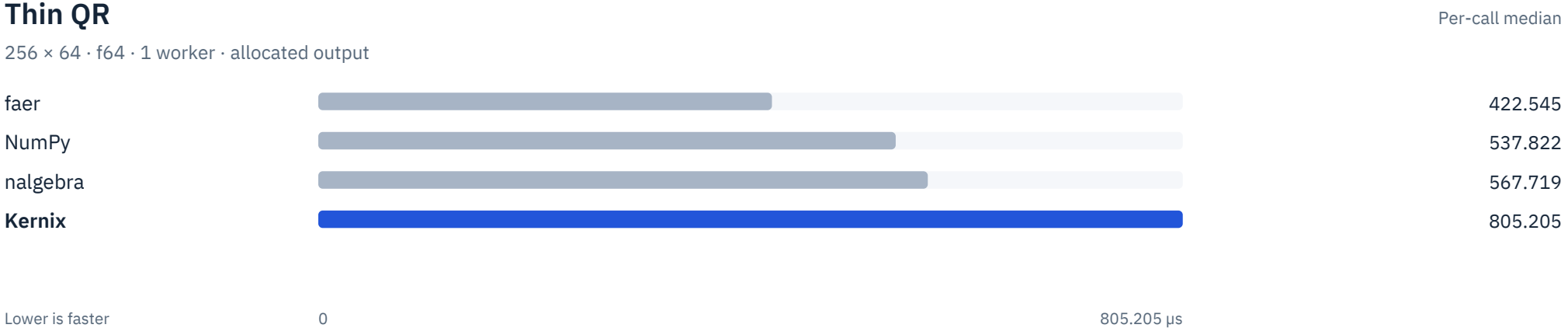
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.8.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

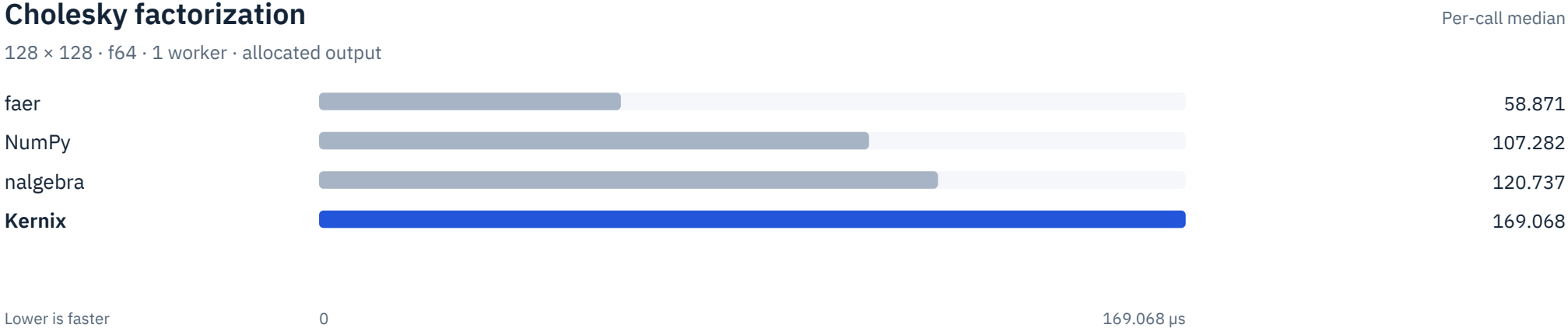
Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.8.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.8.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

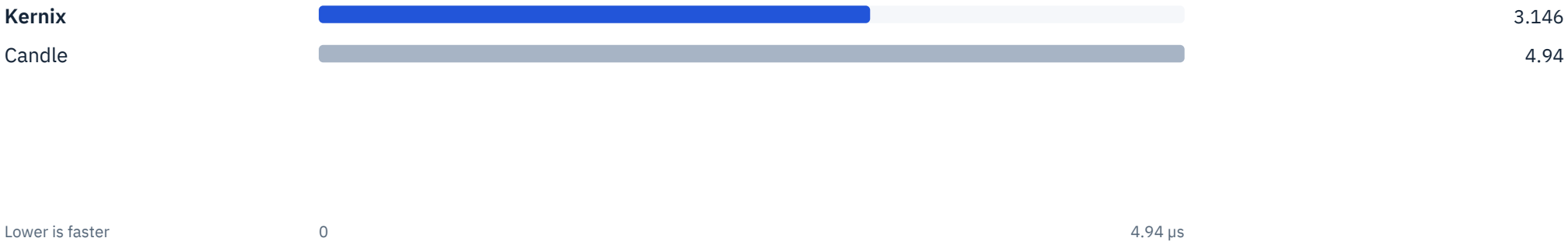
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.8.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented

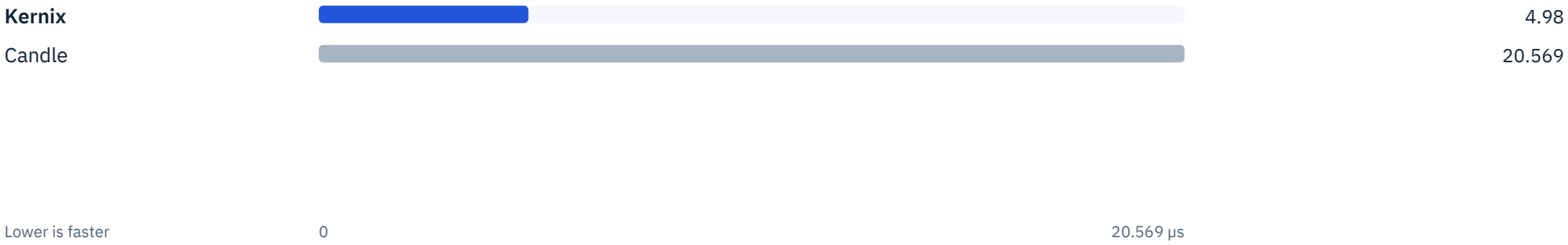
Per-call median



Model backward

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients

Per-call median



Kernix 0.8.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



Kernix 0.8.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

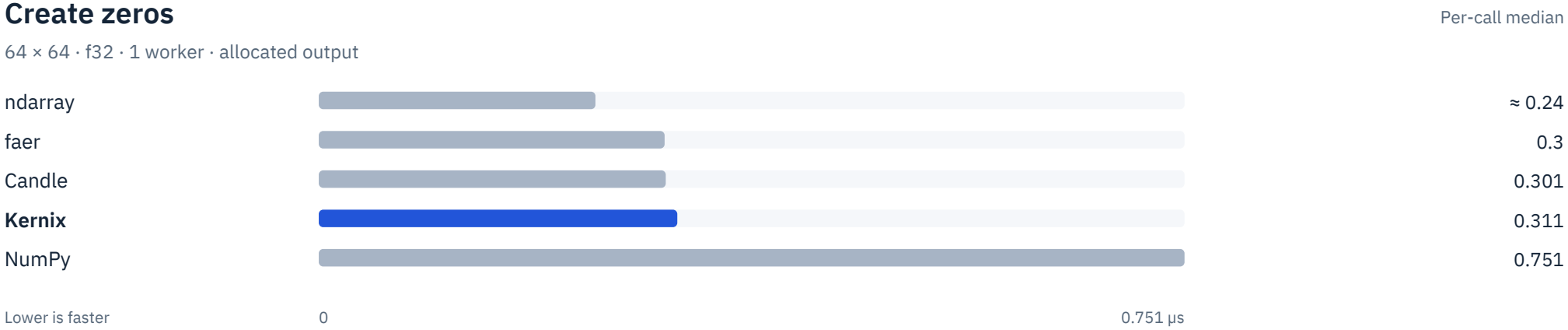


Kernix 0.8.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2525	2.525
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2204	2.204
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2034	2.034
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2174	2.174
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4088	4.088
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2104	2.104
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	14968	14.968
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	14928	14.928
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12744	12.744
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15449	15.449

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13555	13.555
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12854	12.854
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	210025	210.025
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	100529	100.529
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	213732	213.732
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	93687	93.687
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100990	100.99
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	92925	92.925
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	901967	901.967
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	963663	963.663

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	767644	767.644
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	731015	731.015
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	735213	735.213
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	910683	910.683
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7403	7.403
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6592	6.592
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6542	6.542
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8176	8.176
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6523	6.523
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7444	7.444

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	75692	75.692
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	48983	48.983
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	74280	74.28
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	46287	46.287
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	46177	46.177
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	52289	52.289
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	417535	417.535
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	356691	356.691
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	617082	617.082
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	352744	352.744

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	363384	363.384
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	615598	615.598
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3286	3.286
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4198	4.198
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3096	3.096
Thin QR / Kernix 96 × 32 · f64	66746	66.746
Thin QR / faer 96 × 32 · f64	54122	54.122
Thin QR / nalgebra 96 × 32 · f64	57739	57.739
Thin QR / NumPy 96 × 32 · f64	70382	70.382
Thin QR / Kernix 256 × 64 · f64	805205	805.205

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	422545	422.545
Thin QR / nalgebra 256 × 64 · f64	567719	567.719
Thin QR / NumPy 256 × 64 · f64	537822	537.822
Cholesky factorization / nalgebra 64 × 64 · f64	20228	20.228
Cholesky factorization / Kernix 64 × 64 · f64	25808	25.808
Cholesky factorization / faer 64 × 64 · f64	14206	14.206
Cholesky factorization / NumPy 64 × 64 · f64	18174	18.174
Cholesky factorization / NumPy 128 × 128 · f64	107282	107.282
Cholesky factorization / nalgebra 128 × 128 · f64	120737	120.737
Cholesky factorization / faer 128 × 128 · f64	58871	58.871

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	169068	169.068
Cholesky factorization / faer 256 × 256 · f64	316065	316.065
Cholesky factorization / nalgebra 256 × 256 · f64	820354	820.354
Cholesky factorization / Kernix 256 × 256 · f64	1030990.0000000001	1030.9900000000001
Cholesky factorization / NumPy 256 × 256 · f64	1094099	1094.099
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3977	3.977
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7564	7.564
Model inference / Candle batched normalized/sliced MLP · f64	4940	4.94
Model inference / Kernix batched normalized/sliced MLP · f64	3146	3.146
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4087.9999999999995	4.0879999999999995

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Candle batched normalized/sliced MLP · f64	11381	11.381
Model backward / Candle prepared graph → fresh gradients · f64	20569	20.569
Model backward / Kernix prepared graph → fresh gradients · f64	4980	4.98
Parameter update / Kernix transactional SGD, fixed gradients · f64	831	0.831
Row softmax / NumPy 256 × 64 · f64	131027	131.027
Row softmax / Candle 256 × 64 · f64	115458	115.458
Row softmax / Kernix 256 × 64 · f64	120307	120.307
Row softmax / ndarray 256 × 64 · f64	110859	110.859
Create zeros / Kernix 64 × 64 · f32	311	0.311
Create zeros / faer 64 × 64 · f32	300	0.3

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / ndarray 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create zeros / NumPy 64 × 64 · f32	751	0.751
Create zeros / Candle 64 × 64 · f32	301	0.301
Create ones / faer 64 × 64 · f32	391	0.391
Create ones / Kernix 64 × 64 · f32	431	0.431
Create ones / Candle 64 × 64 · f32	461	0.461
Create ones / NumPy 64 × 64 · f32	2054	2.054
Create ones / ndarray 64 × 64 · f32	351	0.351